

SQL Cheat Sheet

Basic Queries

- Select all columns:

This retrieves all columns from the table.

```
SELECT * FROM employees;
```

Example Output:

Output:

```
+-----+-----+-----+
| first_name | last_name | salary |
+-----+-----+-----+
| John      | Doe       | 5000    |
| Jane      | Smith     | 7000    |
+-----+-----+-----+
```

- Select specific columns:

Retrieve only the necessary columns to improve readability and efficiency.

```
SELECT first_name, last_name, salary FROM employees;
```

Example Output:

Output:

```
+-----+-----+-----+
| first_name | last_name | salary |
+-----+-----+-----+
| John      | Doe       | 5000    |
| Jane      | Smith     | 7000    |
+-----+-----+-----+
```

- Filter results using WHERE:

Use WHERE to return only rows that meet specific criteria.

```
SELECT * FROM employees WHERE salary > 8000;
```

Example Output:

Output:

```
+-----+-----+-----+
| first_name | last_name | salary |
+-----+-----+-----+
| Alice      | Brown     | 9000    |
+-----+-----+-----+
```

- Sorting results:

Use ORDER BY to sort results in ascending (default) or descending order.

```
SELECT first_name, last_name, salary FROM employees ORDER BY salary DESC;
```

Example Output:

Output:

```
+-----+-----+-----+
```

first_name	last_name	salary
Alice	Brown	9000
Jane	Smith	7000
John	Doe	5000

String Functions

- Concatenation:

Use || to combine two or more columns into a single string.

```
SELECT first_name || ' ' || last_name AS "Full Name" FROM employees;
```

Example Output:

Output:

Full Name
John Doe
Jane Smith

- Convert to uppercase/lowercase:

Use UPPER() or LOWER() to change text case.

```
SELECT UPPER(first_name), LOWER(last_name) FROM employees;
```

Example Output:

Output:

UPPER_FN	lower_ln
JOHN	doe
JANE	smith

Joins

- Employees and their department:

Use a subquery to match employees with their departments.

```
SELECT first_name, last_name, department_id FROM employees
WHERE department_id IN (SELECT department_id FROM departments);
```

Example Output:

Output:

first_name	last_name	department_id
John	Doe	50

- Employees and their managers:

List employees who share the same manager.

```
SELECT first_name, last_name, manager_id FROM employees
WHERE manager_id IN (SELECT manager_id FROM employees);
```

Example Output:

Output:

+-----+-----+-----+			
first_name	last_name	manager_id	
+-----+-----+-----+			
Jane	Smith	124	
+-----+-----+-----+			

Salary Updates

- Increase salary by 10%:

Use arithmetic operations to update salary values.

```
SELECT first_name, last_name, salary, salary + salary * 0.10 FROM employees;
```

Example Output:

Output:

+-----+-----+-----+			
first_name	last_name	salary	new_salary
+-----+-----+-----+			
John	Doe	5000	5500
+-----+-----+-----+			

- Decrease salary by 200:

Subtract a fixed amount from salaries.

```
SELECT first_name, last_name, salary, salary - 200 FROM employees;
```

Example Output:

Output:

+-----+-----+-----+			
first_name	last_name	salary	new_salary
+-----+-----+-----+			
John	Doe	5000	4800
+-----+-----+-----+			